

```
import java.io.*;
import java.util.*;
```

```
class Todo implements Serializable{
    int id; String t; boolean d;
    Todo(int id,String t){this.id=id; this.t=t;}
    public String toString(){return "["+id+" " +(d?"[x] ":"[ ]")+t;}
}
```

```
public class TodoApp{
    static final String F="todos.ser";
    List<Todo> a=new ArrayList<>(); int next=1;
    Scanner sc=new Scanner(System.in);
    public static void main(String[] args){new TodoApp().go();}
```

```
    void go(){ load();
        for(;;){
            System.out.print("\n1.L 2.A 3.T 4.R 0.X > ");
            switch(sc.nextLine().trim()){
                case "1" -> list();
                case "2" -> add();
                case "3" -> toggle();
                case "4" -> remove();
                case "0" -> { save(); return; }
                default -> System.out.println("?");
            }
        }
    }
    int n(){ try{ return Integer.parseInt(sc.nextLine().trim()); }catch(Exception e){ return -1; } }
```

```
    void list(){ if(a.isEmpty()) System.out.println("(empty)");
        else for(Todo t:a) System.out.println(t); }
    void add(){ System.out.print("Title: ");
        String t=sc.nextLine().trim(); if(t.isEmpty()) return;
        a.add(new Todo(next++, t)); }
    void toggle(){ System.out.print("ID: "); int id=n();
        for(Todo t:a) if(t.id==id){ t.d=!t.d; return; }
        System.out.println("Not found."); }
    void remove(){ System.out.print("ID: "); int id=n();
        a.removeIf(t -> t.id==id); }
```

```
@SuppressWarnings("unchecked")
    void load(){ File f=new File(F); if(!f.exists()) return;
        try(ObjectInputStream in=new ObjectInputStream(new FileInputStream(f))){
            a=(List<Todo>)in.readObject();
            for(Todo t:a) if(t.id>=next) next=t.id+1;
        }catch(Exception e){ System.out.println("Load fail"); } }
    void save(){
        try(ObjectOutputStream out=new ObjectOutputStream(new FileOutputStream(F))){
            out.writeObject(a);
        }catch(IOException e){ System.out.println("Save fail"); } }
}
```